Transfer learning

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Teach reuse through an explicit parameter account.

Use checked threshold, baseline and collision examples while keeping source provenance and measured claims honest.

Keep one habit: Model one case → remove help → assess the different extractor → return to a changed target.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Trace a reused **feature extractor** and a newly fitted **output head**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

No. Identical **input**s to a fixed deterministic head produce identical **output**s. Information lost by the representation cannot be recovered from that representation alone.

Opening retrieval

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate vector length and normalise a nonzero vector

Reference example, not the only acceptable wording: Teacher explanation: Lengths are 5, 10 and 5. Unit vectors are [0.6, 0.8], [0.6, 0.8] and [-0.6, -0.8].

The length difference between Q and A disappears; the opposite direction of C is preserved.

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

Look for: Encode an ordered sequence with an explicit vocabulary

Reference example, not the only acceptable wording: Teacher explanation: IDs are [2, 3, 4, 0], mask [1, 1, 1, 0], real length 3 and final real event STOP at position 2.

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

Look for: Compute max and mean pooling under an explicit window rule

Reference example, not the only acceptable wording: Teacher explanation: Values are 1, 3, 2 and 7. Sum is 13, maximum 7 and mean 3.25. Each pooled **output** has shape 2 by 2.

Pictured end goal

By the end, I can explain **transfer learning** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Similarity and normalisation

Sequence representations

Pooling and spatial information

For the compact alternative schedule, use two 45-minute sessions. Keep three claims separate: real transfer reuses previously learned knowledge; this supplied extractor is a transparent stand-in; the reported scores concern tiny constructed target examples.

No external model, student account or personal dataset is needed. Require students to mark frozen versus fitted quantities before computing.

Outcomes

Trace a reused **feature extractor** and a newly fitted **output head**

Fit a small target-task head using **training** data and compare a baseline

Identify when a frozen representation cannot support a new task

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach the frozen/fitted boundary with checked examples

> Keep the main, pictured and independent extractors distinct.

Definitions, supplied extractor and complete training trace

A source model may learn useful features before we have many labels for a new target task. **Transfer learning** reuses those learned features or parameters. One approach freezes the **feature extractor** and fits a new **output head**.

Fine tuning instead updates some or all of the reused parameters, usually together with the head.

Use a transparent stand-in for a source-trained extractor: f(x) = ReLU(x1 + x2), with supplied weights [1, 1] and bias 0. ReLU keeps a positive value and replaces a negative value with zero.

Here x1 and x2 are nonnegative fictional event counts, so f(x) is their sum. We did not train this extractor on a source dataset; this exercise demonstrates the mechanics of reuse, not measured transfer benefits.

The target is a high-activity label. The extractor returns a single number h, not a class label or probability. A new head converts h into the target **prediction**.

Use target **training** records T1 [1, 1] with label 0, T2 [1, 3] with label 0, T3 [4, 2] with label 1 and T4 [5, 3] with label 1. Their extracted values are 2, 4, 6 and 8.

Fit the head by trying thresholds 3, 5 and 7 and counting **training** mistakes. **Fitting** can use a small search; it need not use gradients.

The tie rule prefers the smaller threshold if **training** mistake counts match; there is no tie here. The extractor remains unchanged. The code assumes finite numeric counts and valid binary labels.

A constant baseline predicts the most frequent **training** label; with two of each, our declared tie rule chooses 0.

Training record · Input counts · Label · Frozen h · Prediction at threshold 5

> **Training record:** T1
>
> **Input counts:** [1, 1]
>
> **Label:** 0
>
> **Frozen h:** 2
>
> **Prediction at threshold 5:** 0

> **Training record:** T2

Input counts: [1, 3]

Label: 0

Frozen h: 4

Prediction at threshold 5: 0

Training record: T3

Input counts: [4, 2]

Label: 1

Frozen h: 6

Prediction at threshold 5: 1

Training record: T4

Input counts: [5, 3]

Label: 1

Frozen h: 8

Prediction at threshold 5: 1

Training record: Candidate threshold 3

Input counts: Training mistakes

Label: 1

Frozen h: Threshold 5: 0

Prediction at threshold 5: Threshold 7: 1

Same-row validation and equality key

Freeze the fitted head before validation

VALIDATION INPUT	FROZEN FEATURE	HEAD AT THRESHOLD 5
V1 [2, 1], label 0	h = 3	Predict 0, correct
V2 [3, 4], label 1	h = 7	Predict 1, correct

The fitted head is correct on 2 of 2 validation records; the constant class 0 baseline is correct on 1 of 2. This tiny constructed comparison is not evidence of general transfer performance.

Guided: h = 5 and **prediction** is 1 because equality is included. The threshold was fitted; weights [1, 1] and bias 0 stayed frozen.

Exit: two constructed validation cases provide limited **evidence**, the extractor was supplied rather than source-trained here, and no from-scratch comparison was run.

Compare both methods on the same target cases with a declared **training** budget and untouched final evaluation after all choices.

Use the same validation records for both methods. These records were not used to fit the threshold. If validation guides later choices, reserve separate untouched final evaluation data.

We have not compared with a full model trained from scratch or measured **training** time, so this result cannot establish either advantage.

A new input [2, 3] uses the fitted threshold 5. Give h and the predicted class, including the equality rule. Which parameters changed during fitting?

Representation collision and supplied repair

New target: first count exceeds second	Supplied input	Correct label	Original frozen h
U	[1, 3]	0	4
V	[3, 1]	1	4
Collision	Same h, conflicting labels	Any deterministic h-only head gives the same output	Cannot classify both correctly
Hand-set repair	h = ReLU(x1 – x2), threshold 1	U → 0; V → 1	Representation changed; no training run claimed

Separate picture givens and interpretation

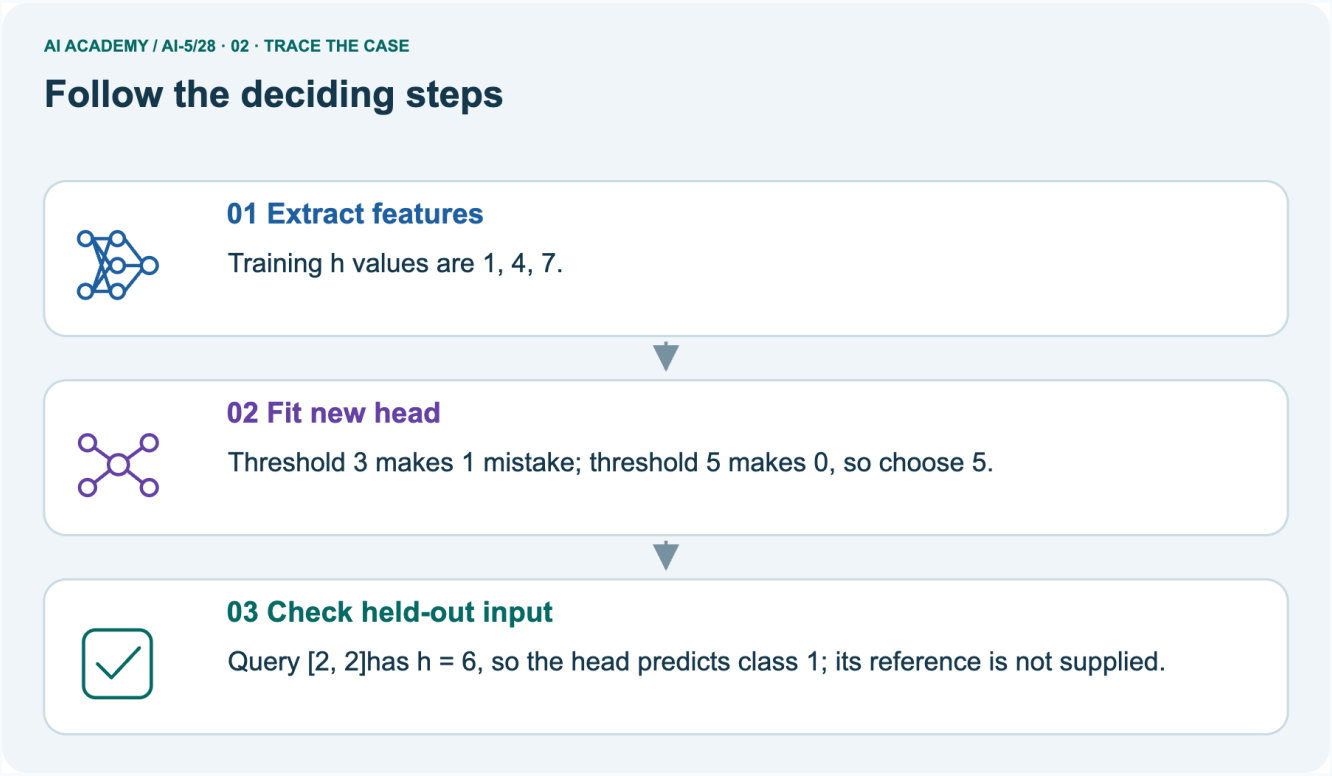
Transfer a fixed **feature extractor** while identifying what is actually refitted.

Frozen extractor h =left+2right. New-task **training input**s [1, 0]label 0,[2, 1]label 0,[1, 3]label 1. Class 1 when h >=threshold; candidates 3 or5.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Keep two extractors separate. The main example freezes h = x1 + x2 and fits threshold 5. Its validation inputs have supplied labels, so correctness can be counted. The picture freezes h = left + 2 × right: training features are 1, 4, 7, and threshold 5 makes zero training mistakes. Its query [2, 2] gives h = 6 and class 1, but has no supplied reference label. A prediction alone cannot be scored as correct.

Picture trace and changed-query key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Extract features	Training h values are 1, 4, 7.
2. Fit new head	Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5.
3. Check held-out input	Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold.

Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

A frozen extractor can miss task-specific distinctions; transfer is a strategy requiring validation, not a guarantee of useful features.

Change: Query changes to [2, 1].

Prediction: Its h becomes 4 and the fitted head predicts class 0.

Reason: Extractor stays fixed; the changed **input** crosses the chosen head's boundary.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Transfer learning	Reusing knowledge learned on a source task or dataset to help a target task
Feature extractor	The part of a model that maps an input to a representation
Output head	The part that maps a representation to the current task's output
Fine tuning	Training some or all previously learned parameters on the target task

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Freeze the fitted head before validation

VALIDATION INPUT	FROZEN FEATURE	HEAD AT THRESHOLD 5
V1 [2, 1], label 0	h = 3	Predict 0, correct
V2 [3, 4], label 1	h = 7	Predict 1, correct

The fitted head is correct on 2 of 2 validation records; the constant class 0 baseline is correct on 1 of 2. This tiny constructed comparison is not evidence of general transfer performance.

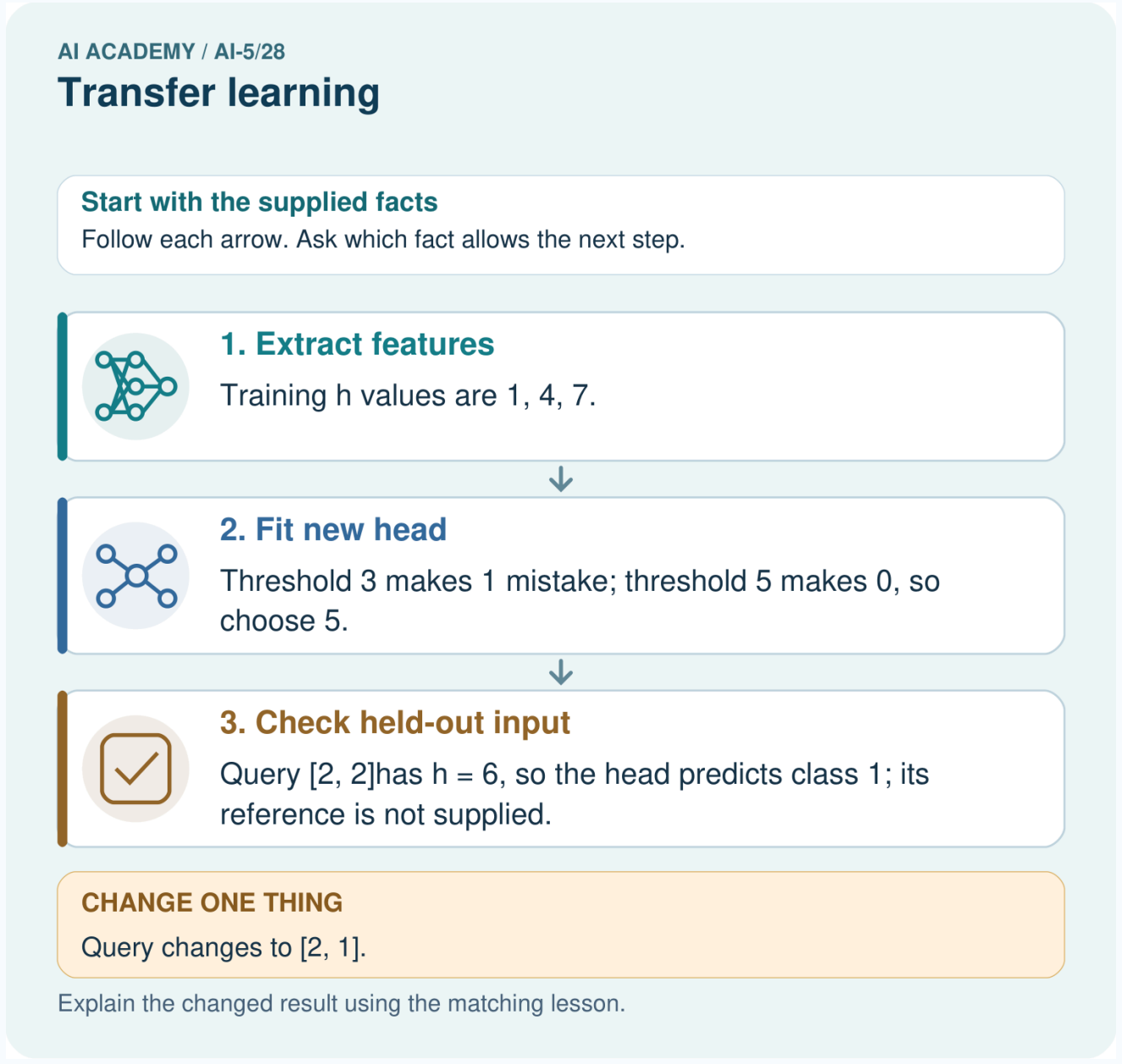
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Transfer learning

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: For the four target **training** records, give the number of mistakes at thresholds 3, 5 and 7 and choose the fitted threshold. Then score that head and the declared constant baseline on V1 and V2.

Explain which data fitted the head.

Teacher check: Teacher explanation: **Training** mistake counts are 1, 0 and 1, selecting threshold 5. Validation **prediction**s are 0 and 1, so the head scores 2 of 2. The constant class 0 baseline scores 1 of 2. Only T1 through T4 fit the head; validation evaluates the fitted choice.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Transfer learning**. Proposed explanation: “Frozen means the whole model cannot learn”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The extractor stays fixed while the new head is fitted on target **training** examples.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Only the threshold head was selected from the new **training** labels. The declared extractor weights [1, 2] were frozen. Changed-case check: Its h becomes 4 and the fitted head predicts class 0.

Extractor stays fixed; the changed **input** crosses the chosen head’s boundary.

Changed-case reasoning: Its h becomes 4 and the fitted head predicts class 0. Extractor stays fixed; the changed **input** crosses the chosen head’s boundary.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide separate **input**, feature, threshold and **prediction** boxes. Allow calculators and mark frozen parameters with a lock symbol on the board.

Extension: on U and V, consider thresholds at or below 4 and thresholds above 4; show that neither case classifies both correctly. Discuss why source performance does not establish target suitability.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the collision principle: a head cannot reconstruct distinctions absent from its input.
7 to 18 minutes	Trace the input, extractor and head. Complete W1 and label frozen weights and fitted threshold.
18 to 32 minutes	Count training mistakes for each threshold. Trace the optional code and declared tie rule.
32 to 45 minutes	Use the validation visual and W2 to compare with the constant baseline on identical records.
Session 2 start to 8 minutes	Complete the guided boundary example and check equality at the threshold.
8 to 20 minutes	Use W3 to expose the new-task collision. Distinguish a hand-chosen repair from fine tuning.
20 to 35 minutes	Collect W4 independently; require the training tie and its declared resolution before validation.
35 to 45 minutes	Use W5 and the exit response to audit evidence. Give feedback, then administer the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main thresholds 3/5/7: mistakes 1/0/1, selected 5. Main validation head 2/2 versus baseline 1/2. Freeze extractor weights and bias; fit only threshold. Independent features 2, 4, 5, 7; thresholds 2/4/6 make 2/1/1 mistakes, tie selects 4. Independent validation features 4 and 2 predict 1 and 0, both correct; baseline class 0 scores 1/2. Require correct equality and training-only selection.

Never infer source-training history or efficiency from these supplied arithmetic cases.

Diagnose the current misconceptions

Frozen means the whole model cannot learn: The extractor stays fixed while the new head is fitted on target **training** examples.

A larger head can undo any lost information: Identical representations remain indistinguishable to a fixed deterministic head, regardless of its size.

Transfer guarantees better or faster results: The source representation may not suit the target. Accuracy and resource benefits require fair measured comparisons.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A separate frozen extractor is h = 2*x1 + x2. **Training input**s [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1.

Compare thresholds 2, 4 and 6 using **prediction** 1 when h >= threshold; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0.

Compare with the **training**-majority baseline, whose tie rule chooses 0.

Teacher explanation: **Training** features are 2, 4, 5 and 7. Mistakes at thresholds 2, 4 and 6 are 2, 1 and 1, so the tie rule selects 4. Validation features are 4 and 2, **prediction**s 1 and 0, scoring 2 of 2. The constant class 0 baseline scores 1 of 2. Equality at threshold 4 gives class 1.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace the reused pipeline

For **input** [4, 2], compute h using the original extractor and the class at threshold 5. Identify the frozen weights and bias, the fitted quantity, and why h is not a probability.

Teacher explanation: The extractor gives h = 6 and the head predicts 1. Weights [1, 1] and bias 0 stay frozen; only the threshold is fitted. h is a nonnegative sum of counts, with no probability mapping or calibration.

W2 Fit and compare

For the four target **training** records, give the number of mistakes at thresholds 3, 5 and 7 and choose the fitted threshold. Then score that head and the declared constant baseline on V1 and V2.

Explain which data fitted the head.

Teacher explanation: **Training** mistake counts are 1, 0 and 1, selecting threshold 5. Validation **prediction**s are 0 and 1, so the head scores 2 of 2. The constant class 0 baseline scores 1 of 2. Only T1 through T4 fit the head; validation evaluates the fitted choice.

W3 Diagnose a lost distinction

For the first-count-greater target, explain why the original extractor cannot distinguish U [1, 3] from V [3, 1]. Trace the hand-chosen [1, -1] extractor and threshold 1.

Does this show that **fine tuning** has been trained and independently validated?

Teacher explanation: Both original features are 4, so a deterministic head must give the same answer despite labels 0 and 1. The changed features are 0 and 2, giving **prediction**s 0 and 1.

This is a hand-chosen change on the examples being inspected, not **training** or independent validation.

W5 Plan credible evidence

A report omits the source checkpoint and pre**process**ing, claims faster learning without timing, and repeatedly chooses settings using the final evaluation set.

Name the missing records and propose a fair comparison with a model trained from scratch.

Teacher explanation: Record source version, provenance and permitted use, pre**process**ing, frozen/trainable parts, splits, baseline and resource budget. Measure time or computation before claiming speed.

Use **training**/validation for choices and new untouched final evaluation after repeated final-set tuning, or report that independent **evidence** is unavailable. Compare both methods on the same target cases and declared budget.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: h = 5 and **prediction** is 1 because equality is included. The threshold was fitted; weights [1, 1] and bias 0 stayed frozen.

Exit: two constructed validation cases provide limited **evidence**, the extractor was supplied rather than source-trained here, and no from-scratch comparison was run.

Compare both methods on the same target cases with a declared **training** budget and untouched final evaluation after all choices.

W1 Trace the reused pipeline

W2 Fit and compare

W3 Diagnose a lost distinction

W4 Fit a new head independently

A separate frozen extractor is h = 2*x1 + x2. **Training input**s [1, 0], [1, 2], [2, 1], [3, 1] have labels 0, 0, 1, 1.

Compare thresholds 2, 4 and 6 using **prediction** 1 when h >= threshold; choose the fewest mistakes, breaking ties toward the smaller threshold. Score it on validation [2, 0] with label 1 and [0, 2] with label 0.

Compare with the **training**-majority baseline, whose tie rule chooses 0.

Teacher explanation: **Training** features are 2, 4, 5 and 7. Mistakes at thresholds 2, 4 and 6 are 2, 1 and 1, so the tie rule selects 4. Validation features are 4 and 2, **prediction**s 1 and 0, scoring 2 of 2. The constant class 0 baseline scores 1 of 2. Equality at threshold 4 gives class 1.
</details>

W5 Plan credible evidence

Independent assessment and fresh retry

W4 requires features 2, 4, 5, 7; mistake counts 2, 1, 1; tie-selected threshold 4; and validation scores 2 of 2 versus 1 of 2.

Fresh retry: frozen h = x1 + 2*x2; **training input**s [0, 1], [2, 1], [1, 2], [1, 3], labels 0, 0, 1, 1. Thresholds 2, 4, 6 again give 2, 1, 1 mistakes, selecting 4.

Validation [0, 2] label 1 and [2, 0] label 0 yield features 4 and 2 and both correct; constant 0 gets one correct. Ask which parameter was fitted.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **transfer learning evidence** sheet to Image-pattern Network Investigation. Show how you can trace a reused **feature extractor** and a newly fitted **output head**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide separate **input**, feature, threshold and **prediction** boxes. Allow calculators and mark frozen parameters with a lock symbol on the board.

Extension: on U and V, consider thresholds at or below 4 and thresholds above 4; show that neither case classifies both correctly. Discuss why source performance does not establish target suitability.

Week 29 studies uncertainty and failure analysis, using specific errors and limitations rather than assuming a confident **output** is correct.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/29, ai-5/31, ai-5/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Trace a reused **feature extractor** and a newly fitted **output head**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
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> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For the four target **training** records, give the number of mistakes at thresholds 3, 5 and 7 and choose the fitted threshold.

Then score that head and the declared constant baseline on V1 and V2. Explain which data fitted the head.

Reasoned practice answer: Teacher explanation: **Training** mistake counts are 1, 0 and 1, selecting threshold 5. Validation **prediction**s are 0 and 1, so the head scores 2 of 2. The constant class 0 baseline scores 1 of 2. Only T1 through T4 fit the head; validation evaluates the fitted choice.

Check every learner before discussion: Use the worked case for **Transfer learning**.
Proposed explanation: “Frozen means the whole model cannot learn”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The extractor stays fixed while the new head is fitted on target **training** examples.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Query changes to [2, 1]. Require a **prediction** and a reason before comparing.
Expected result: Its h becomes 4 and the fitted head predicts class 0.

Reason: Extractor stays fixed; the changed **input** crosses the chosen head's boundary.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

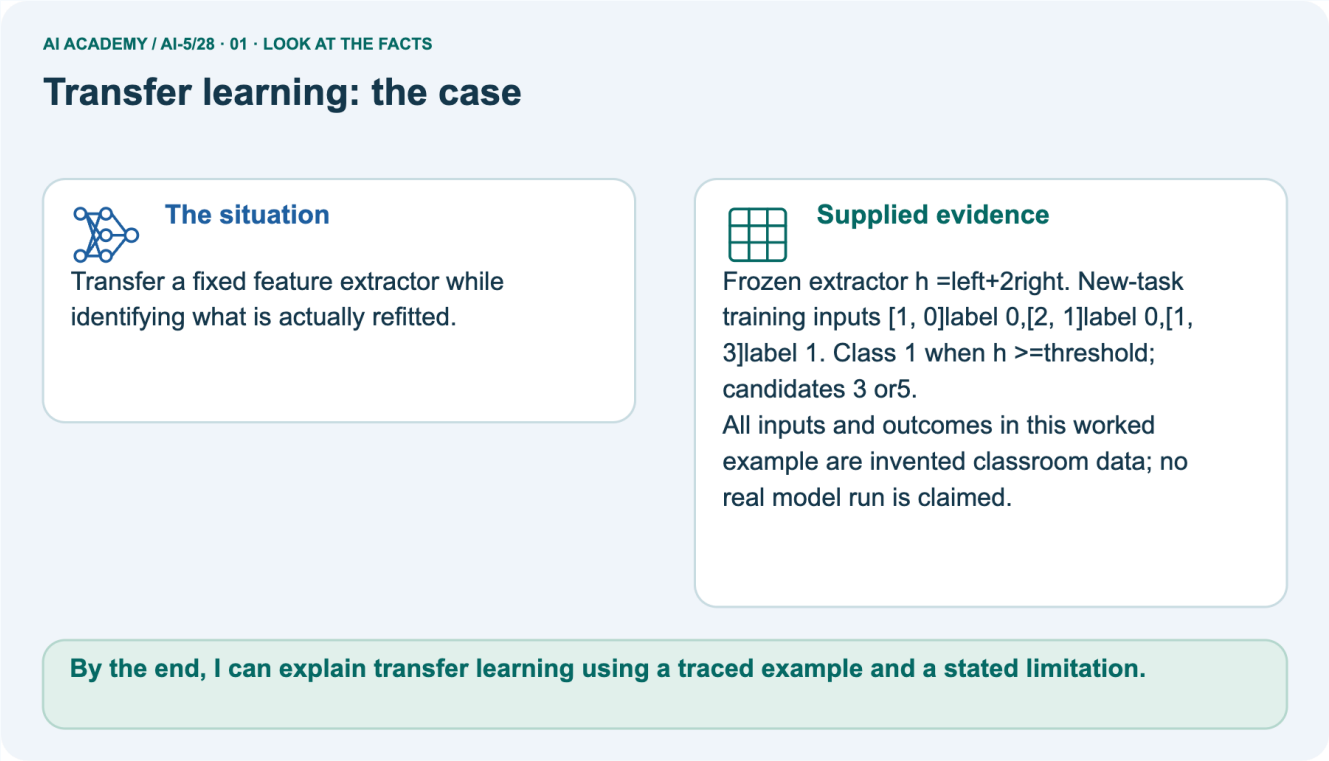
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Transfer learning: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

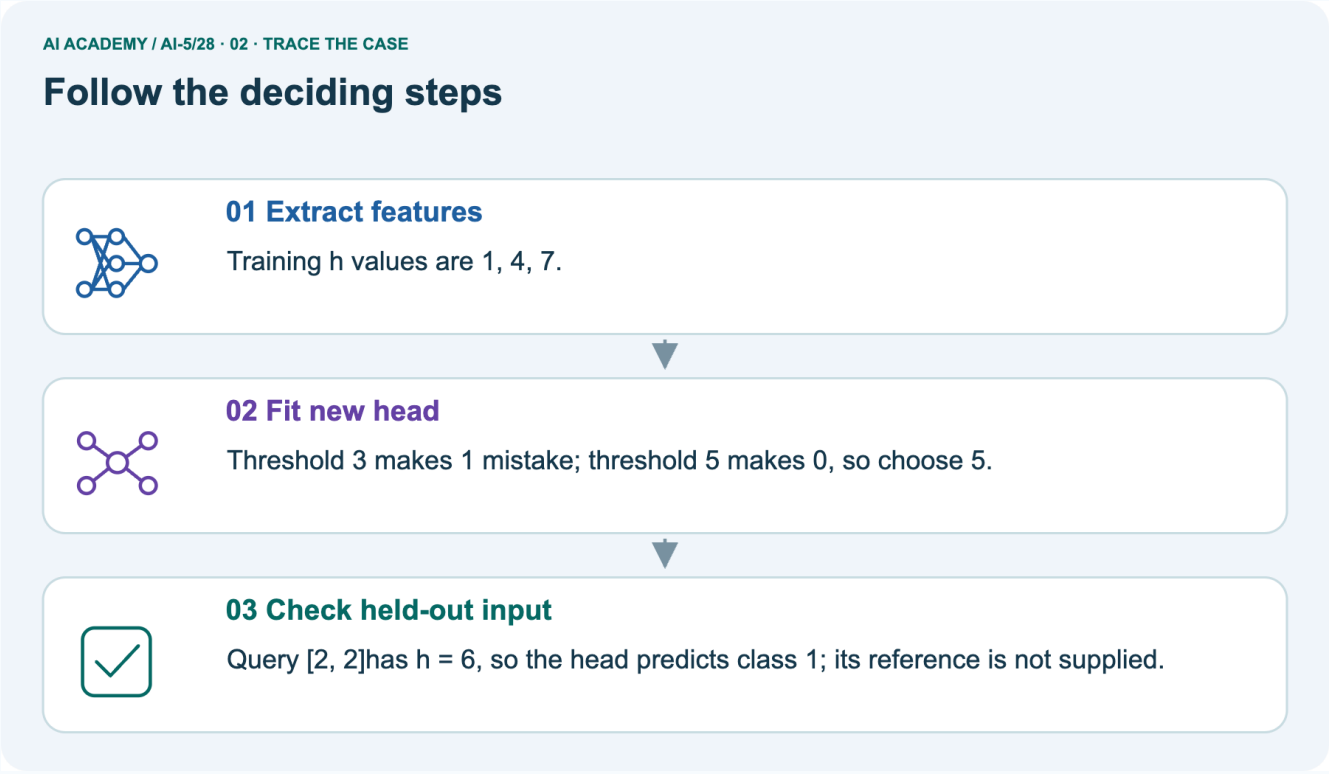
Transfer a fixed **feature extractor** while identifying what is actually refitted. Supplied facts: Frozen extractor h =left+2right. New-task **training input**s [1, 0]label 0,[2, 1]label 0,[1, 3]label 1.

Class 1 when h >=threshold; candidates 3 or5. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



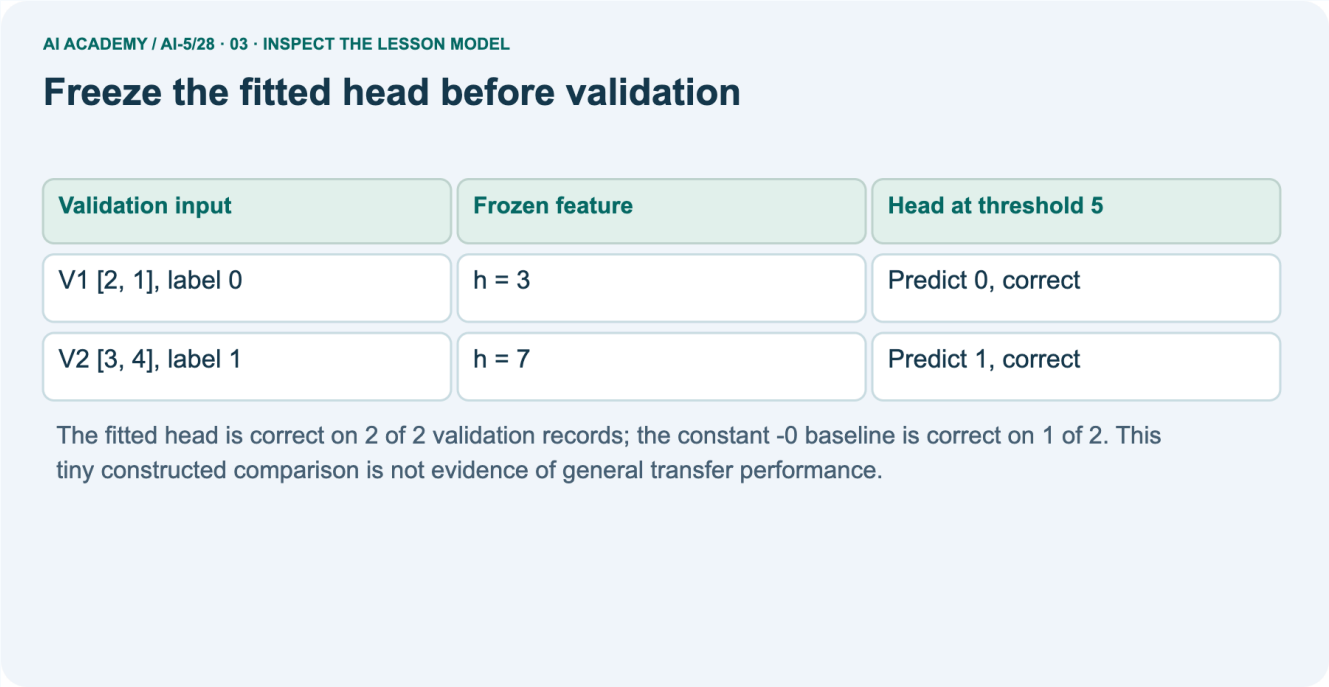
Follow the labelled picture with the matching explanation. Open the vector diagram

Extract features: **Training** h values are 1, 4, 7. → Fit new head: Threshold 3 makes 1 mistake; threshold 5 makes 0, so choose 5. → Check held-out **input**: Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Freeze the fitted head before validation



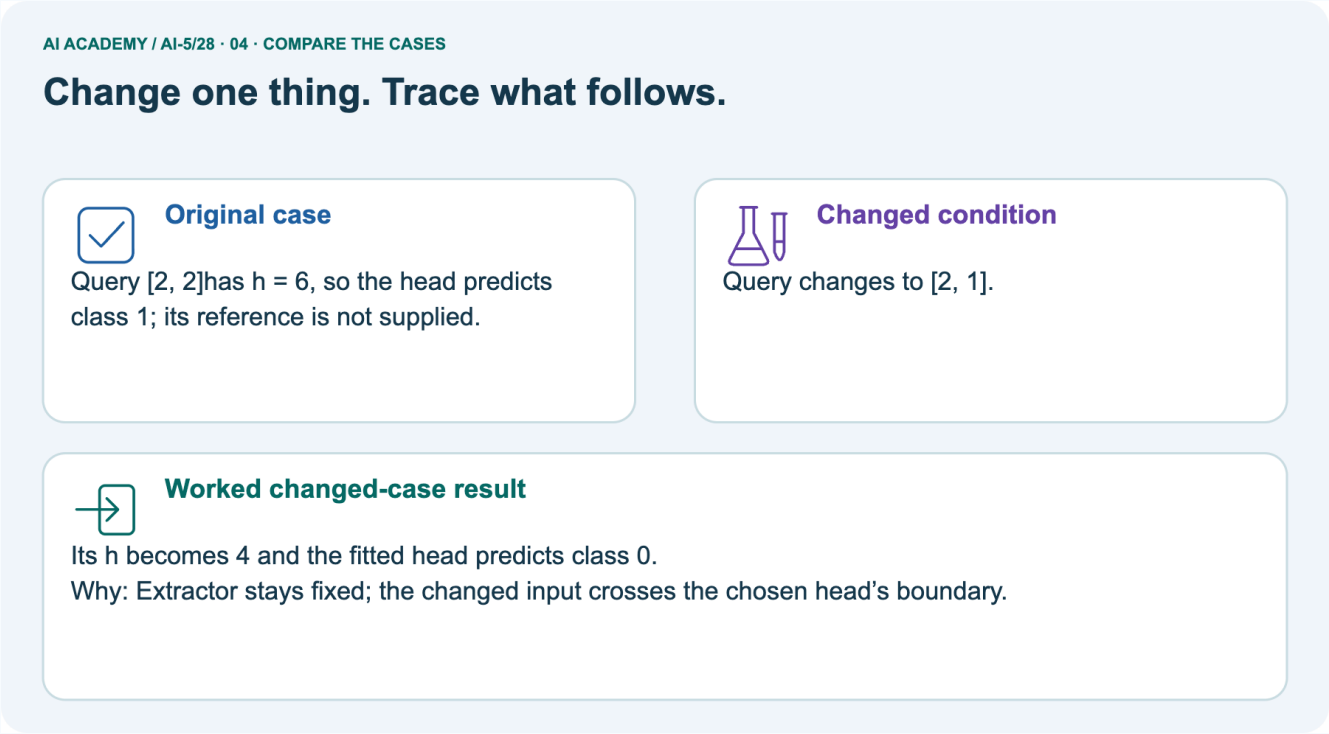
Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze the fitted head before validation. V1 [2, 1], label 0 | h = 3 | Predict 0, correct. V2 [3, 4], label 1 | h = 7 | Predict 1, correct

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Query [2, 2]has h = 6, so the head predicts class 1; its reference is not supplied. Changed condition: Query changes to [2, 1]. Expected result: Its h becomes 4 and the fitted head predicts class 0.

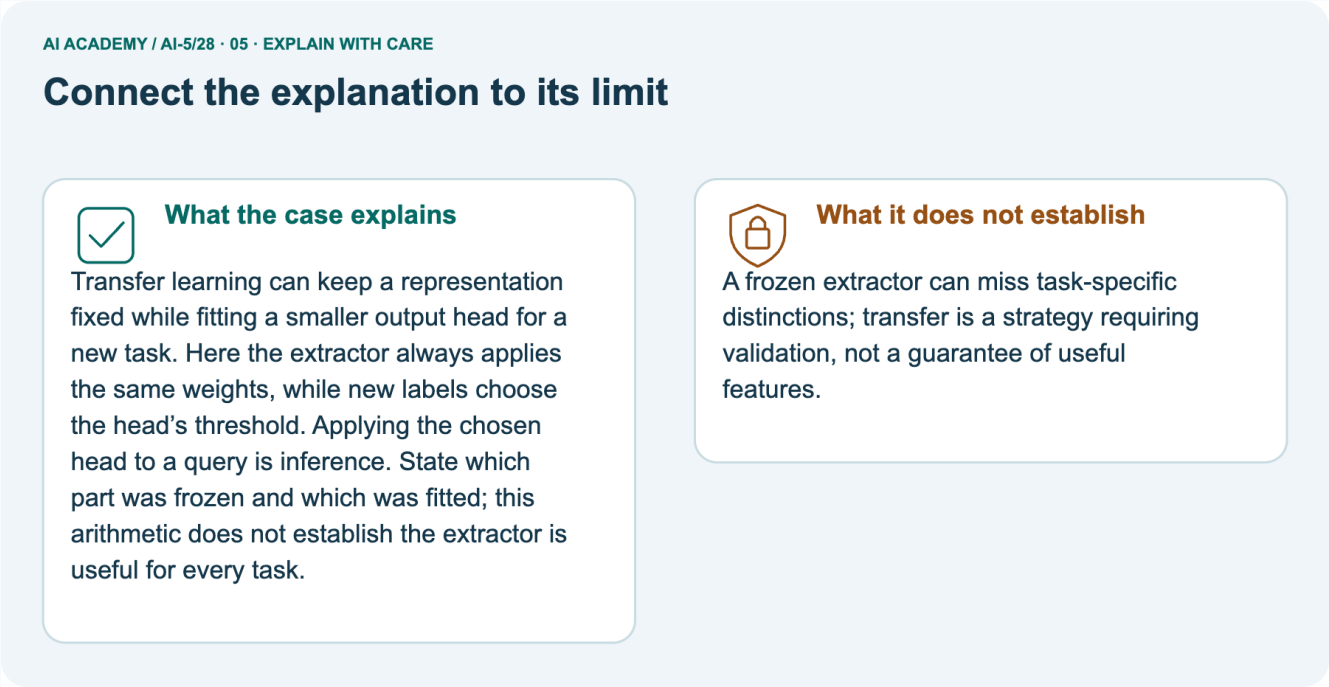
Reason: Extractor stays fixed; the changed **input** crosses the chosen head's boundary.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Extractor stays fixed; the changed **input** crosses the chosen head's boundary.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Transfer learning can keep a representation fixed while **fitting** a smaller **output head** for a new task. Here the extractor always applies the same weights, while new labels choose the head's threshold.

Applying the chosen head to a query is inference. State which part was frozen and which was fitted; this arithmetic does not establish the extractor is useful for every task.

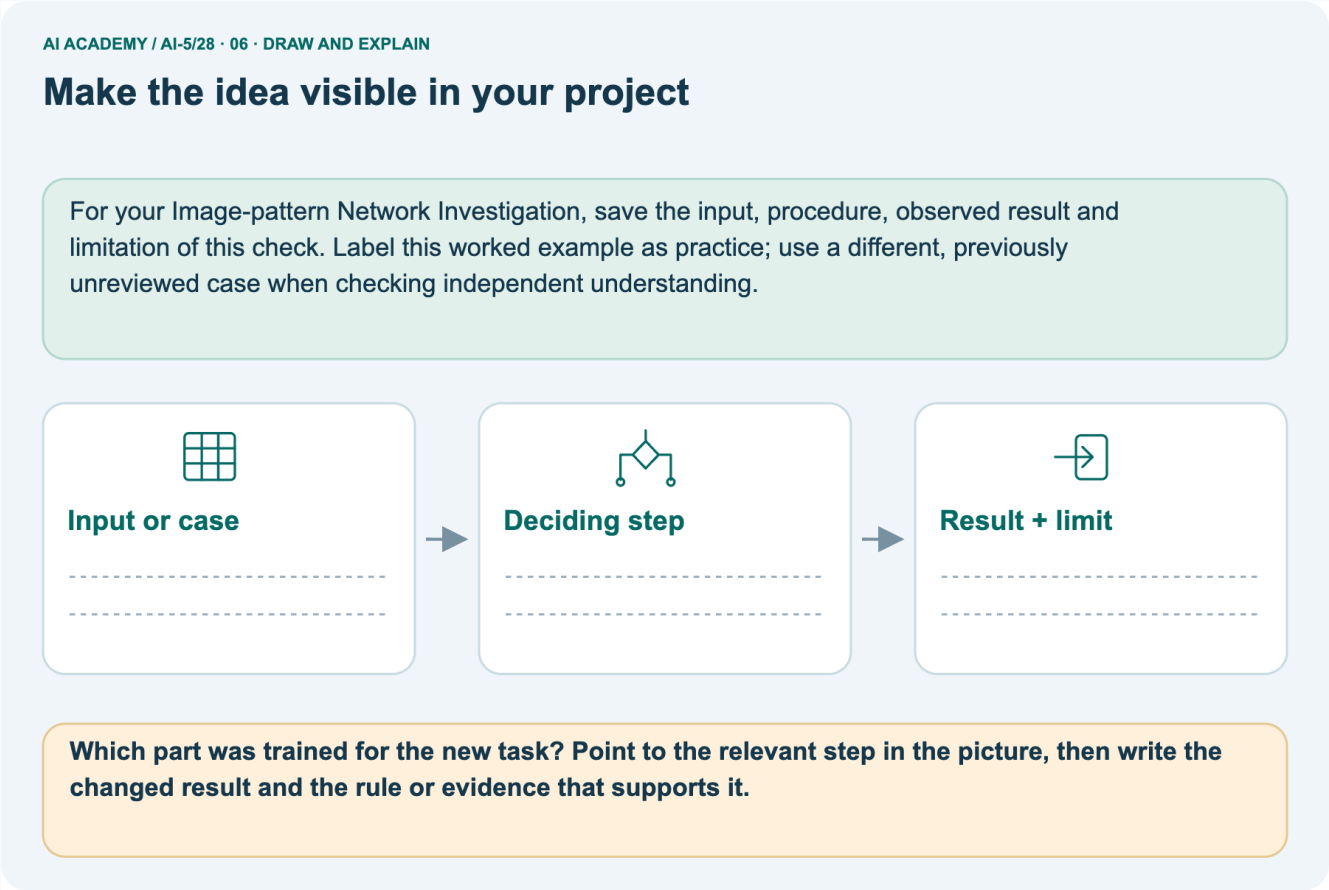
Limit: A frozen extractor can miss task-specific distinctions; transfer is a strategy requiring validation, not a guarantee of useful features.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A frozen extractor can miss task-specific distinctions; transfer is a strategy requiring validation, not a guarantee of useful features.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which part was trained for the new task? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Only the threshold head was selected from the new **training** labels. The declared extractor weights [1, 2] were frozen. Changed-case check: Its h becomes 4 and the fitted head predicts class 0.

Extractor stays fixed; the changed **input** crosses the chosen head’s boundary.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Transfer learning: the case

02. Follow the deciding steps

03. Freeze the fitted head before validation

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.